

Robots for Well-being: Design and Integration of a Low-Cost Social Robot Prototype for Promoting Healthy Habits

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Abstract—Robots have the potential to impact society beyond traditional manufacturing and professional environments positively. Achieving this requires not only technical advancements but also addressing design challenges to tackle social issues effectively. One critical area where robots can make a substantial impact is in ensuring the equitable distribution of high-quality healthcare. However, existing robotic solutions are often costly and inaccessible, limiting their widespread adoption. To address this gap, this work presents Axobotl, a cost-effective and customizable social robot prototype designed to promote healthy habits among students and office workers. Axobotl leverages widely available components and modern hardware, enabling the integration of state-of-the-art AI capabilities. The prototype features a smartphone with a touch display, Dynamixel servo-actuated movements, and a Raspberry Pi-based control system, enabling interactive and expressive behaviors. This paper outlines the design and development process of Axobotl, which follows a human-centered design thinking and participatory design approach, including ideation, definition, and prototyping phases. Axobotl represents an initial step toward democratizing social robotics for preventive healthcare, particularly in developing countries.

Index Terms—Social Robotics, Human-Centered Design, Human-Robot Interaction, Well-being, Preventive Healthcare, Co-design, Participatory Design

I. INTRODUCTION

The World Health Organization (WHO) emphasizes that the primary goal of healthcare is to improve overall health and well-being while ensuring its equitable distribution [1], [2]. However, despite this objective, the prevalence of non-communicable or chronic diseases, such as diabetes, hypertension, cardiovascular diseases, certain types of cancers, sleep apnea, and depression, has been steadily increasing over the past decade and is expected to continue rising [3], [4], particularly in developing countries and regions with high inequality. Tackling this growing health challenge requires a shift from reactive treatment to proactive prevention, focusing on strategies that encourage healthier lifestyle behaviors [1], [5], [6].

Among the populations affected by these challenges are undergraduate students in their final years, graduate students, and academic staff, including professors and administrative personnel. Undergraduate and graduate students often undergo significant life transitions, such as academic pressures and career planning, which can lead to unhealthy habits, including poor sleep, reduced physical activity, and high stress levels. If not addressed early, these behaviors can

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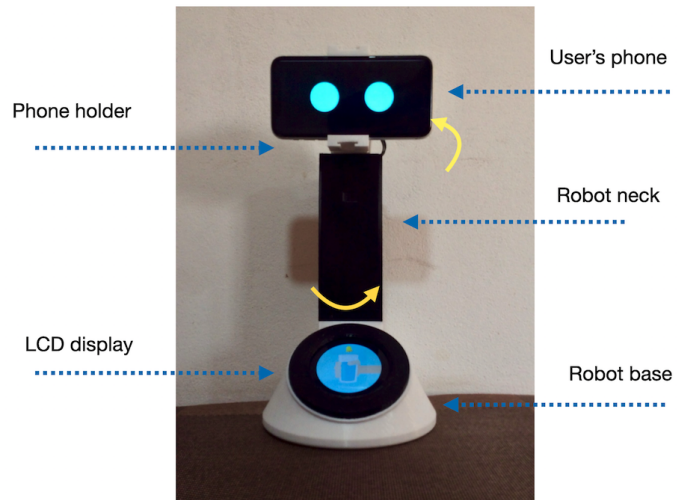


Fig. 1: Initial version of Axobotl, a 3D-printed social desktop robot designed to promote healthy behaviors in undergraduate students and office workers, enabling the integration of AI components

persist into adulthood later, exacerbating long-term health risks [7]. Similarly, academic staff members face comparable challenges, including heavy workloads, long working hours, and elevated professional stress, which can also contribute to poor lifestyle choices and long-term health consequences. Moreover, modern urban and digital lifestyles exacerbate these challenges even further. People of all ages now spend more time on screen-based activities, motorized transportation, and sedentary work [8], [9].

Given the rapid rise of sedentary behaviors, particularly due to screen-based activities and desk jobs, and the limited access to preventive healthcare services, especially in emerging nations, there is an urgent need for accessible and cost-effective health interventions. In countries such as Mexico, which faces considerable health inequalities, the rise of obesity and related chronic diseases underscores the urgency of finding scalable, effective solutions [4], [10]. Traditional healthcare models that rely heavily on reactive treatment are no longer sufficient. To address these growing challenges, it is imperative to shift focus to proactive, prevention-based strategies that can be integrated into the daily lives of students and office workers.

To promote public health and fitness, technologies such as wearable devices and smartphone applications have become increasingly accessible and widely adopted. However, their

effectiveness in driving long-term behavior change remains uncertain [11], [12]. In contrast, social robots offer a more engaging alternative by leveraging physical presence, verbal and non-verbal communication, and multimodal interaction [13]. Unlike smartphone apps, social robots can foster deeper personal connections with users, enhancing engagement, accountability, and motivation [14], [15].

Despite their potential, research on social robots has primarily focused on domains such as eldercare [16], child-robot interaction [17], and rehabilitation [18]. At the same time, their role in promoting healthy behaviors among undergraduate students and office workers remains largely unexplored [19]. Moreover, most social robotic solutions for health promotion are costly and inaccessible to the general population. As highlighted in [20], a significant barrier to the widespread adoption of social robots is their high cost, which limits their availability in diverse social settings, particularly in more unequal regions like Latin America. This highlights the need for robotic solutions that can be tailored to match the resources and needs of users in these regions.

This work represents an initial effort to bridge this research gap by developing Axobotl, an accessible and inclusive robotic prototype. Our objective is to engage undergraduate students in developing robotic systems that address mental and physical health challenges in emerging countries. Axobotl, in this case, is designed to promote a healthier daily lifestyle. To achieve this, we apply a human-centered, design-thinking, and participatory design approach, actively involving undergraduate students as co-designers throughout the various design stages. This article presents the first iteration, detailing the processes of ideation, definition, and prototyping. A depiction of the proposed target scenario and overall design concept is illustrated in Figure 5.

II. RELATED WORK AND CONTRIBUTIONS

The literature includes several studies exploring the use of robotic systems to promote healthy behaviors. However, most of these efforts involve expensive or inaccessible technologies and robotic systems, which are often unavailable to emerging countries, such as wearable robotic exoskeletons [21], industrial robots [22], and humanoid robots [12], [14]. One notable exception is Breakbot [23], a minimally designed, snowman-shaped robot developed to encourage workplace well-being. However, Breakbot relies solely on an Arduino microcontroller for processing, limiting its ability to integrate modern AI-based components. This raises a critical question: How can cost-effective yet advanced robotic systems tailored for developing countries be designed to leverage state-of-the-art AI methods for Human-Robot Interaction?

In this context, two robotics systems designed for use in developing countries stand out. On one hand, the EVA robot [24] was developed to guide therapeutic interventions for individuals with dementia and assist caregivers in managing disruptive behaviors in Latin American countries. On the other hand, the FRED robot [25] was designed to teach computational thinking and support the development of

educational games in Brazilian schools [20]. While these robotics systems were designed to address the research and equity gap in social robotics between developed and developing countries, their application focus and target users differ from those of this project. Therefore, to the best of the authors' knowledge, there is a lack of social robotics solutions tailored to undergraduate and office workers in developing countries, specifically aimed at promoting healthy behaviors.

As an initial step toward addressing this research gap, this work presents the design concept and initial development of a cost-effective, interactive desktop assistant robot denoted as Axobotl. This novel robotic prototype integrates modern hardware components, allowing it to leverage advanced AI capabilities. This enhances its versatility, enabling it to serve various social roles, including acting as an assistant to promote healthy habits. The proposed social robot features a smartphone as its head, mounted on a structure actuated by two Dynamixel servo motors, allowing for simple yet expressive movements. To enhance versatility, a Raspberry Pi 4 serves as the central processing unit, managing motor control, data acquisition, and communication with external devices via Wi-Fi.

The main contributions of this are summarized as follows:

- Description of the design process and concept of a cost-effective desktop assistant robot that utilizes widely available and affordable components. This approach enhances accessibility for students and office workers in developing countries, helping to reduce the technological gap.
- Integration of modern hardware and software technologies to enable advanced AI-based interactions, real-time data processing, and multiple interaction modalities. This includes a smartphone-based display that supports facial expressions and non-verbal cues to encourage healthy habits such as hydration, mobility, and breathing exercises.

III. DESIGN PROCESS

The design process of this prototype represents an initial step towards more advanced human-centered methodologies for creating robotic systems. At this stage, the primary goal is to quickly develop a simple, cost-effective, and functional social robot that undergraduate students, particularly in developing countries like Mexico, can independently build, customize, and experiment with. The design emphasizes simplicity and practicality, demonstrating the potential of social robots enhanced with state-of-the-art AI-based components to address social challenges, specifically targeting undergraduate students. The aim is to create a foundational prototype that can serve as the basis for future co-design workshops and focus group sessions [26], enabling students to iteratively explore, personalize, and develop social robots that promote human well-being and empowerment. This approach aligns with recent advancements in human-centered robotics. For instance, Sienkiewicz et al. [27] employed a participatory design framework involving elderly individuals,

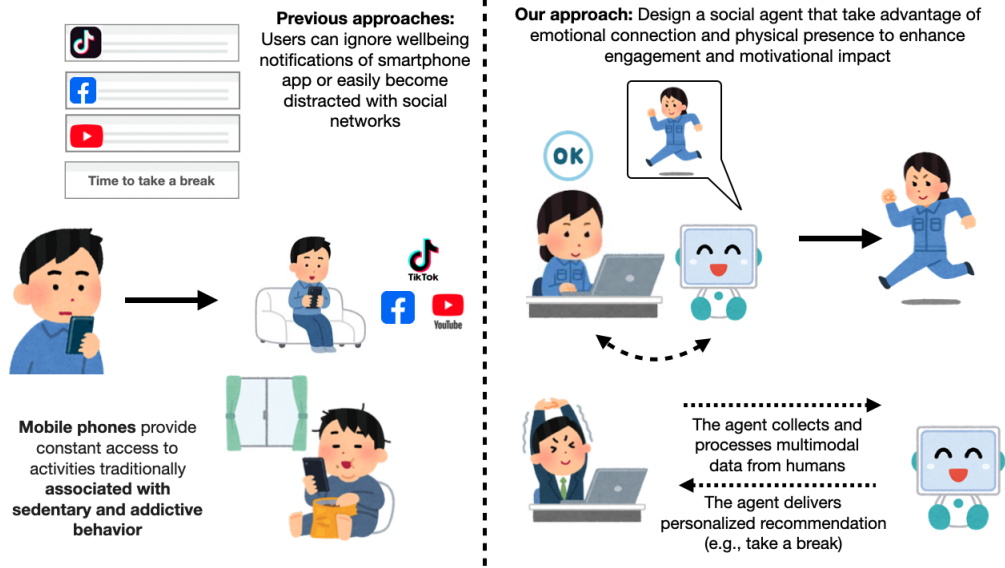


Fig. 2: Design concept of social robots fostering positive impacts on people’s lives by promoting healthy behaviors through physical presence, verbal and non-verbal communication, and multimodal interaction

caregivers, medical professionals, and psychologists to co-create social robots for elder care. Their process incorporated techniques such as card sorting, storyboarding, and semi-structured interviews to ensure the development of empathetic and inclusive robotic solutions. Similarly, Hasnine et al. [28] introduced the Laboratory Safety Assistant (LSA), a socially assistive robot designed through a co-design approach to monitor unsafe behaviors and provide real-time social support for students in laboratory environments. While these studies make significant contributions, introducing new participatory methods, design guidelines, and applications, many remain focused on specific early-stage design phases or rely on commercial robotic platforms. In contrast, this work adopts a holistic approach to developing a fully customizable social robotics platform that empowers students to co-create culturally relevant and context-aware robotic solutions.

The development process was structured into two phases for this initial step. The full process involved a university professor, a human-centered robotics researcher, and two undergraduate students.

A. Phase 1: Define and Ideate

The first phase followed a design thinking methodology that focused on ideation through iterative moments of divergence and convergence [29].

- **Divergence Moments:** Students and researchers engaged in brainstorming sessions or worked individually to generate and sketch concepts.
- **Convergence Moments:** The team discussed, refined, and made key decisions about the robot’s design and functionalities.

The outcome of this phase involved conceptualizing a set of desktop robot design proposals that are able to communicate through various social cues and modalities,

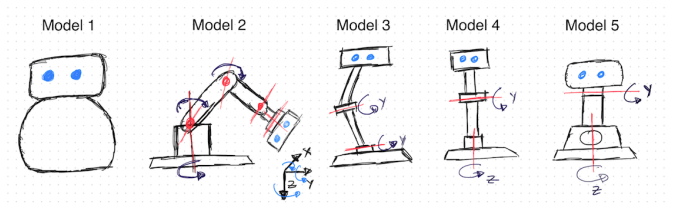


Fig. 3: Five conceptual designs of robot morphologies resulting from divergence moments

such as motions, touch, gestures, and facial expressions. Figure 3 illustrates examples of the conceptual designs or models proposed by researchers and students through convergence sessions. Models 1 and 2 draw inspiration from existing commercial, social, and industrial robots, utilizing a smartphone to display facial expressions. Model 3 features a three-degree-of-freedom design similar to the Yokobo robot proposed in [30], with modifications to link lengths and the kinematic model, which are also applied in Models 3 and 4. To simplify the design while maintaining engaging motion capabilities, Model 5 reduces the degrees of freedom to two. Additionally, a base display was introduced to enable interaction even when the smartphone is not attached to the robot.

Building upon this initial concept, we identified two key functionalities:

- **Improve productivity of the user:** The robot should serve as a holder and charging dock for the user’s smartphone, helping users stay focused and productive.
- **Provide healthier habit motivation:** Through multimodal interactions, including touchscreen prompts, facial expressions, and non-verbal behaviors, the robot must act as both a reminder and motivator for users to

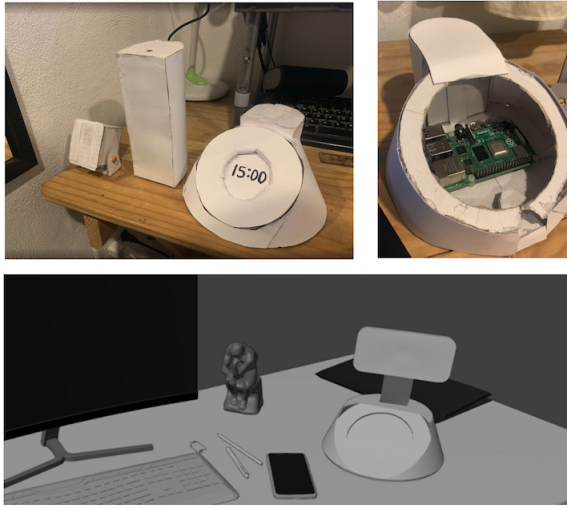


Fig. 4: Example of paper-based prototypes and 3D simulations developed for undergraduate students used to explore diverse scenarios and designs

adopt healthier habits.

To visualize potential interactions and scenarios, the team developed a paper-based prototype to explore various design possibilities. Additionally, 3D simulations were created using Blender to model and evaluate scenarios before moving on to physical prototyping (Figure 4).

B. Phase 2: Functional Prototype Development

The second phase focused on refining the initial prototype, defining its functional and non-functional requirements, performance criteria, and interaction modes. Key aspects such as shape and technology selection were considered in parallel. Two undergraduate students were responsible for implementing the motorized movements, hardware, and software architecture. The final functional, non-functional, and performance requirements are outlined below:

1) Functional Requirements

- **Smartphone Support:** The robot must securely hold a smartphone as its “head” or display without obstructing the screen.
- **Tilting Movement:** The robot must be able to tilt the phone holder forward and backward, simulating a nodding gesture.
- **Rotating Base:** The base must allow rotation around the vertical axis to orient the robot toward the user or different directions.
- **Configurable Motor Setup:** The design must support two different motor configurations to optimize energy efficiency and ease of implementation.

2) Non-Functional Requirements

- **Stability & Safety:** The base must be stable enough to prevent tipping and ensure the phone’s security.
- **Compact Size:** The robot should be small enough to fit on a desk without occupying excessive space.
- **Aesthetic Appeal:** The design must be visually appealing and suitable for a study or work environment.

- **Ease of Construction:** The design must be simple and easy to assemble using accessible materials and tools.
- **3D Printing Compatibility:** All structural components must be designed for 3D printing to facilitate reproducibility and assembly.

3) Performance Criteria

- **Tilting Range:** The phone holder should tilt at least 30° forward and 15° backward.
- **Rotation Range:** The base must support at least 180° rotation around the vertical axis.
- **Load Capacity:** The robot must support a smartphone weighing up to 250g.
- **Smooth Movements:** Both tilting and rotation must be fluid and controlled, avoiding abrupt or jerky motions.
- **Positioning Accuracy:** The motors should allow precise positioning with a tolerance of $\pm 2^\circ$.

Figure 6 illustrates an example of an interaction between Axobotl and its target users, undergraduate students. This interactive scenario showcases Axobotl’s role in promoting healthy habits through engaging and intuitive exchanges.

IV. SYSTEM ARCHITECTURE

Figure 7 illustrates the general system architecture of the Axobotl prototype. This architecture extends and adapts the Yokobo system architecture for integrating multimodal inputs, enabling motor control, and connecting with smart devices, as described in [30], [31]. The central processing unit is a Raspberry Pi 4, which facilitates communication via TCP/IP connections through the NEP+ framework [32]. This framework, based on ZeroMQ, provides a lightweight, cross-platform, and high-performance solution for seamless integration between diverse systems, including advanced robotics platforms such as the Robot Operating System (ROS), game engines, and smart devices [33]. Following a similar approach to the Yokobo robot [30], NEP+ is employed to connect internal components and external smart devices (e.g., smartphones) [31]. While the current iteration does not require ROS, future versions could incorporate advanced robotics algorithms, such as behavior trees for high-level planning. Motor control is achieved through Dynamixel actuators, which are managed via TTL/RS-485 signals using the U2D2 PHB Set and U2D2 interface. This interface connects to the Raspberry Pi through USB-Serial communication. Additionally, a touch display is integrated using SPI/I2C connections to the Raspberry Pi 4. Table I displays the prices of the main components. The minimum cost for this initial prototype is approximately \$426, which is significantly lower than most popular social robots with advanced connectivity capabilities used in research, such as Nao (\$12,990), Pepper (\$32,000), and Jibo (\$825). However, optimizing servo motor selection based on the required torque, as detailed in the next section, will lead to further cost reductions.

V. MECHANICAL DESIGN

The mechanical design consists of 3D-printed structural components made from PETG. The robot consists of four links, including a fixed base, a rotating neck section, the



Fig. 5: Paper-based and 3D-printed prototypes of the initial version of the proposed social robot



Fig. 6: Example interaction between Axobotl and an undergraduate student

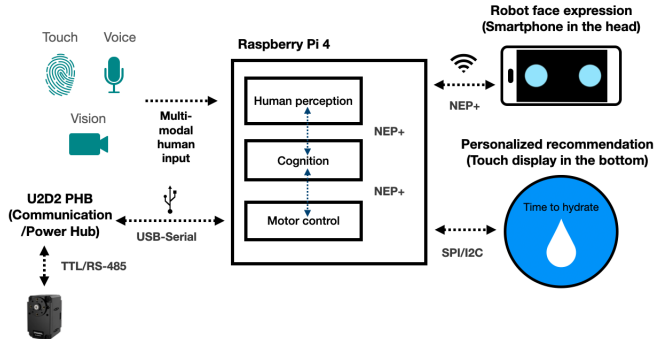


Fig. 7: Axobotl general system architecture concept based and extended from Yokobo [30], [31]

neck itself, and a phone holder. It features two rotational joints: one at the neck, enabling rotation around the vertical (Z) axis, and another at the phone holder, allowing forward and backward tilting. The mobile base has a diameter of 12 cm and a height of 2 cm, designed to house a Dynamixel MX-28 AR servo motor for rotation. The base comprises a fixed part that rests on the desk and holds the servo motor, and a rotating part connected to the servo's shaft, supporting the neck and the rest of the structure. The vertical neck is a 15 cm hollow cylinder with a 2 cm diameter, allowing

Component	Price
Dynamixel MX-28 AR	\$250
Dynamixel XL430-W250	\$50
Raspberry Pi 4	\$35
U2D2 PHB Set	\$19
U2D2 Interface	\$32
TFT touch display	\$40

TABLE I: List of main and most expensive components with their current prices. Due to the modular design, the smartphone is not included in the cost of the robot, as it is assumed that users already have a smartphone.

cable routing from the base to the upper joint. The upper joint houses the Dynamixel XL430-W250-T servo motor for inclination, providing a connection between the neck and the phone holder while ensuring structural robustness for smooth movements. The phone holder is adjustable for widths between 6 cm and 9 cm, securing the phone without obstructing the screen and connecting directly to the inclination servo's shaft.

A. Torque Calculation for Phone Holder Inclination

The torque required for the phone holder inclination is determined based on the combined mass of the phone (0.25 kg) and its support (0.158 kg), the gravitational acceleration (9.81 m/s^2), and the moment arm distance (0.05 m). The calculated torque is:

$$\tau_{\text{inclination}} = (0.25 + 0.158) \times 9.81 \times 0.05 = 0.2001 \text{ Nm}$$

Applying a safety factor of 2:

$$\tau_{\text{inclination, safe}} = 0.4002 \text{ Nm} \approx 4.08 \text{ kg} \cdot \text{cm}$$

B. Torque Calculation for Neck Rotation

The upper section of the robot, including the neck, joints, phone holder, and phone, has a total mass of 0.5253 kg. The torque required for neck rotation is computed using the rotational friction coefficient (0.1) and the neck radius (0.015 m):

$$\tau_{\text{rotation}} = 0.5253 \times 9.81 \times 0.015 \times 0.1 = 0.00773 \text{ Nm}$$

Applying a safety factor of 2:

$$\tau_{\text{rotation, safe}} = 0.01546 \text{ Nm} \approx 0.158 \text{ kg} \cdot \text{cm}$$

C. Optimizing Servo Motor Selection

The initial motor selection was based on the Yokobo robot, utilizing a Dynamixel XL430-W250-T for the phone holder inclination and a Dynamixel MX-28 AR for neck rotation. The Dynamixel XL430-W250-T, with a maximum torque of 1.5 Nm, exceeds the required 0.4002 Nm for the phone holder inclination, ensuring stable and reliable operation. Similarly, the Dynamixel MX-28 AR, which provides 0.39 Nm of torque, comfortably meets the 0.01546 Nm requirement for neck rotation, allowing smooth movement. However, to reduce costs in future design iterations, alternative Dynamixel motors with lower torque ratings will be evaluated while maintaining functionality and performance. For neck rotation, the XL330-M288-T (0.78 Nm) is a potential candidate. Meanwhile, the Dynamixel XL-320 is a suitable choice for the phone holder inclination due to its compact size and affordability. By utilizing both the XL330-M288-T (25\$) and the Dynamixel XL-320 (25\$), the cost of the robot's main electronic components can be reduced to \$176.

D. Minimum Base Diameter for Stability

To maintain stability at a maximum tilt angle of 15° , the minimum base diameter is determined using the center of gravity height (0.1815 m):

$$d_{\text{base}} = 2 \times 0.1815 \times \tan(0.2618) = 0.0972 \text{ m}$$

With a safety factor of 1.5:

$$d_{\text{base, safe}} = 0.1459 \text{ m}$$

Thus, the robot requires a minimum base diameter of approximately 0.1459 m to ensure stability during operation.

VI. EVALUATION OF MECHANICAL, THERMAL, AND STRUCTURAL PERFORMANCE

This section presents the results of the system's performance evaluation, focusing on tilting and rotational movements, structural stability, and thermal behavior of the control unit. The experimental results were analyzed in relation to the design requirements described in the previous section.

1) PWM and Motion Stability

Motor control via Pulse-Width Modulation (PWM) demonstrated high sensitivity in the system's mechanical response. In initial tests, a PWM value of 150 was found to produce smooth and stable movements, particularly for lightweight structures. Values below this threshold led to abrupt or overly slow movements.

With a 250 g load attached to the phone holder, the PWM value of 150 yielded acceptable performance, though with a delayed response. Increasing the PWM to 250 resulted in

significantly improved maneuverability, producing fluid and secure motion. However, when a 540 g load was applied, the motor exhibited torque loss and reduced responsiveness, indicating that the system is not designed to handle such loads while maintaining stability and control.

These results partially validate the system's load capacity performance criterion, confirming that 250 g is a reasonable operational limit.

2) Lower Motor Movement (Rotation)

The lower motor, responsible for rotating the phone holder, completed a full 360° rotation in 5.56 seconds. This satisfies the requirement of achieving at least 180° of rotation. However, residual vibration was observed at the end of the motion, suggesting the need for a progressive braking system to mitigate inertial spikes that may affect stability and positioning accuracy.

3) Structural Stability

External perturbation tests revealed slight wobbling in the upper structure, though the robot was able to recover due to the base's width. This behavior satisfies the RNF1 requirement on stability and safety. Nonetheless, it is recommended to increase friction at the base, for instance by adding rubber pads, to improve surface grip and reduce unintentional displacement or vibration.

4) Thermal Considerations of the Control System

During prolonged operation, the Raspberry Pi 4 exhibited high-temperature warnings. This behavior is attributed to the confined enclosure and absence of active ventilation. Although this does not directly impact functional requirements, it is crucial for long-term operation in desktop environments. To ensure safe temperatures during extended sessions, passive or active cooling components such as heat sinks or ventilation systems should be incorporated.

VII. CONCLUSIONS AND FUTURE WORK

This work presented the design process of Axobotl, a social robot developed to promote healthy habits among students and office workers. The process encompassed ideation, definition, and prototyping, with a focus on making social robotics more accessible for preventive healthcare, particularly in underserved communities. While Axobotl represents a promising step in this direction, the research is still in its early stages, and several limitations need to be addressed. The design process incorporated undergraduate students throughout the ideation, definition, and prototyping phases, informing both the robot's form and function. However, a key limitation is the lack of formal testing or evaluation with the broader target user base. The current version has not yet undergone structured validation in terms of usability, engagement, or effectiveness in behavior change. To address this, future work will involve participatory design activities, including co-design workshops and iterative user feedback sessions, to refine Axobotl's features and ensure cultural and contextual relevance. Additionally, real-world user studies will assess its impact on health behavior, usability, and user acceptance.

Beyond its technical aspects, this project advances accessibility in robotics, particularly in developing countries. While still in its early stages, Axobotl represents a promising step toward inclusive, context-aware social robotics. It has the potential to serve as both a foundation and an inspiration for future applications in preventive healthcare, especially in settings where affordable and adaptable technologies can create a meaningful impact.

REFERENCES

- [1] U. A. Betz, L. Arora, R. A. Assal, H. Azevedo, J. Baldwin, M. S. Becker, S. Bostock, V. Cheng, T. Egle, N. Ferrari *et al.*, “Game changers in science and technology-now and beyond,” *Technological Forecasting and Social Change*, vol. 193, p. 122588, 2023.
- [2] T. A. Ghebreyesus, H. Fore, Y. Birtanov, and Z. Jakab, “Primary health care for the 21st century, universal health coverage, and the sustainable development goals,” *The Lancet*, vol. 392, no. 10156, pp. 1371–1372, 2018.
- [3] S. Ling, K. Brown, J. K. Miksza, L. M. Howells, A. Morrison, E. Issa, T. Yates, K. Khunti, M. J. Davies, and F. Zaccardi, “Risk of cancer incidence and mortality associated with diabetes: A systematic review with trend analysis of 203 cohorts,” *Nutrition, Metabolism and Cardiovascular Diseases*, vol. 31, no. 1, pp. 14–22, 2021.
- [4] R. Gutiérrez Tolentino, I. Lazarevich, M. A. Gómez Martínez, J. A. Barriguete Meléndez, B. Schettino Bermúdez, J. J. Pérez González, R. del Muro Delgado, and C. C. Radilla Vázquez, “Epidemiological overview of overweight and obesity related to eating habits, physical activity and the concurrent presence of depression and anxiety in adolescents from high schools in Mexico city: A cross-sectional study,” in *Healthcare*, vol. 12, no. 6. MDPI, 2024, p. 604.
- [5] P. Grant, *Proactive Care*. Cham: Springer Nature Switzerland, 2024, pp. 87–107. [Online]. Available: https://doi.org/10.1007/978-3-031-69944-3_7
- [6] Y. Wang, L. Wu, J.-P. Lange, A. Fadhil, and H. Reiterer, “Persuasive technology in reducing prolonged sedentary behavior at work: A systematic review,” *Smart Health*, vol. 7, pp. 19–30, 2018.
- [7] S. Puente-Hidalgo, C. Prada-García, J. A. Benítez-Andrades, and E. Fernández-Martínez, “Promotion of healthy habits in university students: Literature review,” in *Healthcare*, vol. 12, no. 10. MDPI, 2024, p. 993.
- [8] S. J. Biddle, I. Petrolini, and N. Pearson, “Interventions designed to reduce sedentary behaviours in young people: a review of reviews,” *British journal of sports medicine*, vol. 48, no. 3, pp. 182–186, 2014.
- [9] V. Cabanas-Sánchez, L. García-Cervantes, L. Esteban-Gonzalo, M. J. Girela-Rejón, J. Castro-Piñero, and O. L. Veiga, “Social correlates of sedentary behavior in young people: The up&down study,” *Journal of Sport and Health Science*, vol. 9, no. 2, pp. 189–196, 2020.
- [10] M. Aceves-Martins, L. López-Cruz, M. García-Botello, Y. Y. Gutierrez-Gómez, and C. F. Moreno-García, “Interventions to prevent obesity in Mexican children and adolescents: systematic review,” *Prevention Science*, pp. 1–24, 2022.
- [11] A. N. Sullivan and M. E. Lachman, “Behavior change with fitness technology in sedentary adults: a review of the evidence for increasing physical activity,” *Frontiers in public health*, vol. 4, p. 289, 2017.
- [12] N. T. Fitter, M. Mohan, R. C. Preston, M. J. Johnson, and K. J. Kuchenbecker, “How should robots exercise with people? robot-mediated exergames win with music, social analogues, and gameplay clarity,” *Frontiers in Robotics and AI*, vol. 10, p. 1155837, 2024.
- [13] K. Kim, L. Boelling, S. Haesler, J. Bailenson, G. Bruder, and G. F. Welch, “Does a digital assistant need a body? the influence of visual embodiment and social behavior on the perception of intelligent virtual agents in AR,” in *2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 2018, pp. 105–114.
- [14] E. Coronado, T. Shinya, and G. Venture, “Hold my hand: Development of a force controller and system architecture for joint walking with a companion robot,” *Sensors*, vol. 23, no. 12, p. 5692, 2023.
- [15] I. Leite, G. Castellano, A. Pereira, C. Martinho, and A. Paiva, “Empathic robots for long-term interaction: evaluating social presence, engagement and perceived support in children,” *International Journal of Social Robotics*, vol. 6, pp. 329–341, 2014.
- [16] D. Deuff, I. Milleville-Pennel, I. Ocnareescu, D. Garcin, C. Aznar, S. Cappy, S. Hagane, P. F. Osorio Marin, E. Coronado Zuniga, L. Rincon Ardila *et al.*, “Together alone, Yōkobo, a sensible presence robot for the home of newly retired couples,” in *Proceedings of the 2022 ACM Designing Interactive Systems Conference*, 2022, pp. 1773–1787.
- [17] E. Coronado, X. Indurkha, and G. Venture, “Robots meet children, development of semi-autonomous control systems for children-robot interaction in the wild,” in *2019 IEEE 4th international conference on advanced robotics and mechatronics (ICARM)*. IEEE, 2019, pp. 360–365.
- [18] T. She and F. Ren, “Enhance the language ability of humanoid robot nao through deep learning to interact with autistic children,” *Electronics*, vol. 10, no. 19, p. 2393, 2021.
- [19] I. Damen, H. Brombacher, C. Lallemand, R. Brankaert, A. Brombacher, P. Van Wesemael, and S. Vos, “A scoping review of digital tools to reduce sedentary behavior or increase physical activity in knowledge workers,” *International journal of environmental research and public health*, vol. 17, no. 2, p. 499, 2020.
- [20] J. Favela, D. Cruz-Sandoval, M. M. d. Rocha, and D. C. Muchaluat-Saade, “Social robots for healthcare and education in Latin America,” *Communications of the ACM*, vol. 67, no. 8, pp. 70–71, 2024.
- [21] Y. Ren, S. H. Kang, H.-S. Park, Y.-N. Wu, and L.-Q. Zhang, “Developing a multi-joint upper limb exoskeleton robot for diagnosis, therapy, and outcome evaluation in neurorehabilitation,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 21, no. 3, pp. 490–499, 2012.
- [22] S. Kashi and S. Levy-Tzedek, “Smooth leader or sharp follower? playing the mirror game with a robot,” *Restorative neurology and neuroscience*, vol. 36, no. 2, pp. 147–159, 2018.
- [23] S. Reeder, L. Kelly, B. Kechavarzi, and S. Sabanovic, “Breakbot: a social motivator for the workplace,” in *Proceedings of the 8th ACM Conference on Designing Interactive Systems*, 2010, pp. 61–64.
- [24] M. M. da Rocha, D. Cruz-Sandoval, J. Favela, and D. C. Muchaluat-Saade, “An open-source socially assistive robot for multisensory healthcare therapies,” in *Workshop on Multisensory Experiences (SensoryX)*. SBC, 2022.
- [25] M. M. D. Rocha and D. C. Muchaluat-Saade, “Friendly robot for education and healthcare: Fred,” in *Proceedings of the 2023 ACM International Conference on Interactive Media Experiences Workshops*, 2023, pp. 19–22.
- [26] D. Kulic, E. Coronado, P. Carreno-Medrano, M. Mintrom, G. Venture, S. Sumartojo, and L. Tian, “User expectations of robots in public spaces: A co-design methodology,” (*No Title*), 2020.
- [27] B. Sienkiewicz, Z. Radosz-Knawa, and B. Indurkha, “A participatory design approach to using social robots for elderly care,” in *2024 IEEE International Conference on Advanced Robotics and Its Social Impacts (ARSO)*. IEEE, 2024, pp. 194–199.
- [28] M. N. Hasnine, B. Indurkha, and M. M. H. Ahmed, “Socially assistive robot as laboratory safety assistant for science students,” in *2024 IEEE International Conference on Advanced Robotics and Its Social Impacts (ARSO)*. IEEE, 2024, pp. 37–42.
- [29] D. Deuff, I. C. Ocnareescu, E. Coronado, L. K. R. Ardila, I. Milleville-Pennel, and G. Venture, “Designing a way of thinking in a robotics research project,” *Journal of the Robotics Society of Japan*, 2020.
- [30] S. Cappy, P. Osorio, S. Hagane, C. Aznar, D. Garcin, E. Coronado, D. Deuff, I. Ocnareescu, I. Milleville, and G. Venture, “Yōkobo: a robot to strengthen links amongst users with non-verbal behaviours,” *Machines*, vol. 10, no. 8, p. 708, 2022.
- [31] S. Cappy, E. Coronado, P. Osorio, S. Hagane, D. Deuff, and G. Venture, “Integration of a presence robot in a smart home,” in *2023 3rd International Conference on Computer, Control and Robotics (ICCCR)*. IEEE, 2023, pp. 192–197.
- [32] E. Coronado, N. Yamanobe, and G. Venture, “Nep+: A human-centered framework for inclusive human-machine interaction development,” *Sensors*, vol. 23, no. 22, p. 9136, 2023.
- [33] E. Coronado, T. Ueshiba, and I. G. Ramirez-Alpizar, “A path to industry 5.0 digital twins for human-robot collaboration by bridging nep+ and ros,” *Robotics*, vol. 13, no. 2, p. 28, 2024.